

Predicting Contract Renewal in Public Digital Procurement

A Survival-Analysis Approach on BOAMP Notices for Gigalis
Internship Report — Executive Orientation

Internship Report — Gigalis (Pays de la Loire)

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Abstract

This is the **executive report** of the internship: it lets a reader scan the project rapidly and understand *what was done*, *why it matters*, and *what is reliable*. The full implementation and diagnostics are in the **technical report** (`phase1_technical_report.pdf`); the data reliability and event-definition detail are in the **data quality report**.

The work concerns predictive analysis of public digital procurement for **Gigalis**, a central purchasing body in Pays de la Loire. The goal is to move from a descriptive view (“what happened?”) towards a predictive one (“when will a contract be re-tendered?”). Because BOAMP contains *no explicit renewal field*, the renewal relationship is reconstructed from observable signals: a proxy renewal **event** built from Sentence-Transformer text embeddings combined with CPV, temporal, and buyer signals, used as the event variable for survival analysis. On 1,100 eligible source contracts the linking method recovers a plausible renewal for 705 of them (**64.1%**), against an 11.3% lexical baseline. Survival modeling gives a global Kaplan–Meier median survival of **48.2 months**, a Cox concordance of **0.6541**, and a best-fitting log-normal parametric model (AIC 7,176.6). The fitted model is translated into 12- and 24-month renewal-risk indicators at contract, buyer, and segment level. Buyer identities were consolidated by a SIREN enrichment step (completed 2026-06-24). The headline caveat: **event** is a proxy, and its precision has not yet been validated by manual review.

1 Context and Objective

1.1 Why this project

Gigalis operates a central purchasing body for digital services in Pays de la Loire. To anticipate re-tendering workloads, it wants to predict *when* digital contracts come up for renewal. The core obstacle: BOAMP publishes calls for tenders and awards, but *no notice says “this renews contract X”*. The renewal link must be inferred from text, classification, timing, and buyer identity.

1.2 Objective

The internship objective is to build a transparent, reproducible pipeline that (i) reconstructs a proxy renewal **event** from BOAMP signals, (ii) uses it as the outcome of a survival model of contract lifetime, and (iii) translates the model into operational renewal-risk indicators. This report orients the reader to that pipeline and its main results; exact field logic, formulas, and diagnostics are deferred to the technical and data quality reports.

2 Data Sources and Scope

2.1 Scope

- **Sectors:** CPV divisions 48 (software), 72 (IT services), 32 (telecom), 35 (security).
- **Territory:** departments 44, 49, 53, 72, 85 (Pays de la Loire).
- **Period:** 2015–2024. Study end date: 2024-12-31.
- **Modeling path:** BOAMP only. DECP is documented but not linked (no shared key between the two sources).

2.2 Corpus numbers

- 3,181 BOAMP notices downloaded (2015–2024).
- 1,933 are APPEL_OFFRE (source contracts); 1,081 are ATTRIBUTION.
- 1,100 are *eligible*: their expected renewal window falls before 2024-12-31.
- 833 are structurally censored upfront (too recent to observe a renewal).

DECP (3,039 digital contracts, same scope) is far richer (100% SIRET, 99.4% durations) but cannot be merged to BOAMP at contract level; contract-level DECP merging remains a future path.

2.3 Key data quality issues (modeling consequences)

Table 1: Key BOAMP data-quality findings and their modeling consequence. Full field-level detail is in the data quality report.

Finding	Value	Consequence
Buyer SIRET coverage (raw)	~9% of corpus	Grouping uses normalised names; SIREN enrichment later consolidates many (see §3.2).
Amount fill rate (APPEL_OFFRE)	~21%	Amount excluded from linking signals.
Declared duration granularity	69% at 12/24/36/48 mo	Duration used as a covariate, not as survival target.
Imputed durations	25% of eligible cohort	Flagged with <code>dur_was_imputed</code> ; imputation is 48 months.
No explicit renewal label	structural	<code>event</code> is a proxy reconstructed by the linking algorithm.

2.4 Taxonomy

Each contract is tagged with one of eleven technology categories (Software & Applications, IT Services & Consulting, Telecom & Networks, Cybersecurity, Cloud & Infrastructure, Data & AI, etc.) via a **deterministic rule-based mapping** from CPV codes and description keywords. This is *not* a trained classifier — the planned NLP classifier was not implemented. Output: `data/processed/taxonomy.csv`.

3 Main Pipeline

The pipeline runs end to end from raw BOAMP notices to operational risk indicators. Table 2 summarises each component and where its outputs live; the subsections below describe what each stage does, and Section 4 reports the numbers.

Table 2: Component status and key outputs. Status: **Done**, **Partial**, **Not started**.

Component	Status	Key result	Where
Data cleaning & EDA	Done	3,181 notices → 1,933 source contracts	data/processed/boamp_full_clean.csv
Buyer SIREN enrichment	Done	660/1,100 eligible rows on a SIREN key	buyer_siren_enrichment/
Renewal linking (event reconstruction)	Done	705/1,100 eligible linked (64.1%)	boamp_renewal_linking_quality/
Survival dataset construction	Done	1,100 rows × 36 cols	data/processed/boamp_phase2_survival.csv
Kaplan–Meier & log-rank	Done	Median 48.2 mo; duration significant	notebooks/02_survival_modeling_boamp.ipynb
Cox PH model	Done	C-index 0.6541	same notebook + reports/tables/survival/
Parametric AFT models	Done	Log-normal best (AIC 7,176.6)	same notebook
Sensitivity analysis	Done	Ranking stable across 3 definitions	reports/tables/survival/sensitivity_comparison.csv
Robustness checks	Done	Placebo, bias check, threshold sweep	validation_robustness/
Operational risk indicators	Done	12/24-mo risk at contract/buyer/segment	reports/tables/survival/top20_renewal_risk.csv
Manual event validation	Partial	Sample drawn; review not done	event_validation/
Trained NLP classifier	Not done	Taxonomy is rule-based only	—
Change-point detection (Phase 4)	Not done	Scoped but not implemented	—

3.1 BOAMP cleaning

Raw notices are parsed from the BOAMP API, deduplicated, and standardised: buyer key construction, contract start-date refinement (using `ATtribution` back-references where available), duration cleaning and imputation (missing eligible durations set to 48 months, flagged), CPV normalisation, and text normalisation. Output: `data/processed/boamp_full_clean.csv`.

3.2 Buyer SIREN enrichment

Because raw SIRET coverage is weak (~9% of the corpus), buyers were initially grouped by normalised name, which fragments a single public entity across spelling variants. A **SIREN enrichment step** (completed **2026-06-24**) queries the official *API Recherche d'Entreprises* for each buyer name and, for **HIGH-confidence matches only**, upgrades the buyer key from `NAME:` to `SIREN:{9-digit}`, merging name aliases of the same legal entity. Of the 1,100 eligible contracts, **660 carry a SIREN key and 440 remain name-keyed**; 275 of 525 unique buyers (52.4%) matched with HIGH confidence. Medium/low matches are kept for review but never applied automatically. Enrichment is a buyer-resolution improvement, *not* ground truth: rare false merges at HIGH confidence remain possible. Outputs: `buyer_siren_enrichment/` and `data/processed/boamp_phase2_survival_siren_enriched.csv`.

3.3 Renewal-event reconstruction

Each eligible source contract is matched against later BOAMP notices from the same buyer using a four-signal composite score: semantic text embeddings (Sentence-Transformer, `paraphrase-multilingual-MiniLM-L12-v2`), CPV hierarchy match, temporal proximity, and buyer-identity agreement (weights 0.40 / 0.25 / 0.20 / 0.15). The best candidate passing the hard text-similarity filter is selected as the renewal successor and sets `event = 1`; contracts with no qualifying successor before study end are right-censored (`event = 0`). See technical report §5 for the full formula.

3.4 Survival modeling

The survival dataset (`data/processed/boamp_phase2_survival.csv`) has one row per eligible contract with observed/censored duration and covariates (`declared_duration_months`, `dur_was_imputed`, `start_year`, category, procedure type). Three model families were fitted in `notebooks/02_survival_modeling_boamp.ipynb`: Kaplan–Meier (with log-rank tests), a Cox proportional-hazards model, and parametric AFT models (Weibull, log-logistic, log-normal). The fitted model was stress-tested with a sensitivity analysis over three event definitions and a battery of robustness checks.

3.5 Operational risk indicators

The best parametric model (log-normal AFT) is applied to all scored contracts to produce conditional renewal probabilities at 12 and 24 months ahead, aggregated to contract, buyer, and segment level. These are prioritisation indicators, not legally verified renewal probabilities. Output: `reports/tables/survival/top20_renewal_risk.csv`.

4 Main Results

4.1 Renewal linking

Table 3: Renewal-linking outcome.

Quantity	Value	
Source contracts (<code>APPEL_OFFRE</code>)	1,933	
Eligible (renewal window observable)	1,100	
Linked renewals (<code>event</code> = 1)	705	(64.1%)
Right-censored (<code>event</code> = 0)	395	(35.9%)
Lexical baseline linking rate	11.3%	

Linking rate by category (Table 4): Software & Applications and IT Services & Consulting renew most (71.3%, 67.9%); Data & AI and IT Hardware & Equipment least (38.2%, 47.6%). Small categories ($n < 10$) are underpowered.

Table 4: Event rate by technology category (1,100 eligible contracts; events sum to 705).

Category	n	events	rate %
IT Services & Consulting	312	212	67.9
Software & Applications	209	149	71.3
Telecom & Networks	204	124	60.8
Cybersecurity	121	75	62.0
Unknown	108	60	55.6
Digital Workplace & Collaboration	45	30	66.7
Data & AI	34	13	38.2
IT Maintenance & Support	22	15	68.2
IT Hardware & Equipment	21	10	47.6
Cloud & Infrastructure	19	13	68.4
GIS & Mapping	5	4	80.0

Link quality. Of the 705 accepted links: 133 are single-candidate (no competing match); of the remaining 572 multi-candidate links, 265 (46%) have a score margin below 0.05, flagging an ambiguous winner. 100 links (14.2%) satisfy the `high_confidence_strict` criterion ($S \geq 0.70$ and margin ≥ 0.05 , multi-candidate only); these form the Model C core (Table 7). **Main unlinked failure mode:** 331 of 395 unlinked contracts (83.8%, rounded 84%) have no temporal partner in the search window — a true absence of a candidate notice, not a scoring failure.

4.2 Survival analysis

Kaplan–Meier. Global median survival is **48.2 months**, consistent with the prevalence of 48-month framework contracts. Log-rank tests (Table 5): renewal timing is driven by declared duration and imputation status ($p < 0.001$), not by technology category ($p = 0.182$).

Table 5: Log-rank tests for differences in renewal timing.

Stratifying variable	statistic	p	df
Category (multivariate)	6.23	0.182	4
Duration was imputed	24.58	< 0.001	1
Declared-duration group	62.30	< 0.001	2
Procedure type (collapsed)	2.06	0.357	2

Cox proportional-hazards model. In-sample concordance: **0.6541**. Two predictors are significant (Table 6): longer declared duration lowers hazard (HR 0.991 per month), and later start year raises it (HR 1.094 per year). No category or procedure effect. The positive `start_year` hazard is partly structural: 2021–2024 contracts are right-censored because a typical 48-month contract would renew after the 2024-12-31 study close.

Table 6: Cox model: significant covariates (multivariate).

Covariate	HR	95% CI	p
<code>declared_duration_months</code>	0.991	[0.987, 0.995]	< 0.001
<code>start_year</code>	1.094	[1.050, 1.140]	< 0.001
<code>dur_was_imputed</code>	0.864	[0.718, 1.040]	0.123
Category / procedure terms	not significant ($p > 0.1$)		

Parametric AFT models. Log-normal fits best (AIC 7,176.6, concordance 0.672), outperforming log-logistic (7,248.8) and Weibull (7,380.3). Full results: `reports/tables/survival/parametric_aic_comparison.csv`.

Sensitivity to event definition. Three event definitions were tested (Table 7). The Cox concordance is stable across all three (0.654, 0.645, 0.633), confirming that the *ranking* of renewal risk is robust even under the strictest definition, even though the absolute event rate and KM level move with the threshold.

Table 7: Sensitivity to event definition. “—” = KM median never reached. KM median in months.

Variant	events	rate %	KM median	Cox C
A — baseline (all links)	705	64.1	48.2	0.654
B — conservative (≥ 0.50)	497	45.2	52.8	0.645
C — strict H.C. ($S \geq 0.70$, $\Delta \geq 0.05$, multi-cand.)	100	9.1	—	0.633

Proportional-hazards assumption. The Schoenfeld test flags a violation for `declared_duration_months` ($p < 10^{-5}$); all other covariates pass. Stratification or a time interaction for duration is future work. The log-normal AFT avoids the PH assumption and gives the same conclusions.

4.3 Operational risk indicators

Table 8: Operational risk summary (1,097 scored contracts).

Metric	Value
Mean 12-month renewal probability	0.190
Mean 24-month renewal probability	0.332
Max 12-month probability (any contract)	0.435
Contracts in <i>medium</i> band (0.25–0.50)	222
Contracts in <i>low</i> band (< 0.25)	875
Expected renewals within 12 months	208
Expected renewals within 24 months	365
Top buyer by expected 12-month volume	Pays de la Loire (18.6)
Top segment by expected 12-month volume	IT Services & Consulting (61.6)

5 Reliability and Limitations

5.1 Reliability evidence

The event variable was stress-tested with checks in `validation_robustness/`:

- **Threshold sensitivity (7 scenarios, score 0.35 to 0.90):** Cox concordance stays above 0.53 throughout (it dips to 0.535 only at the extreme ≥ 0.80 cut); directional findings stable. Full table: `validation_robustness/outputs/threshold_sensitivity_summary.csv`.
- **Placebo check:** winning-link mean score 0.587 vs. runner-up 0.439 vs. synthetic no-buyer 0.434 — the scoring discriminates (21.0% of winners reach ≥ 0.70 vs. 1.2% of runner-ups).
- **Event/censoring bias:** SMD = 0.13 (duration), -0.03 (start year) between event and censored groups — covariates balanced; the dominant bias is structural right-censoring of 2022–2024 contracts.
- **Manual validation:** 150-row sample drawn (`event_validation/outputs/manual_validation_sample.csv`), protocol documented. **Record-level review not yet executed;** precision/recall of the linking step therefore unknown.

5.2 Key limitations

- **Proxy event.** `event = 1` means a plausible renewal link was found, not a legally certified renewal. By composite score, 29.5% of the 705 links are below 0.50 (low), 49.5% between 0.50 and 0.70 (medium), and 21.0% at or above 0.70 (high).
- **Residual buyer fragmentation.** SIREN enrichment consolidated 660 of the 1,100 eligible rows onto a SIREN key, but 440 remain name-keyed, so spelling variants not resolved by the API can still create false negatives. Enrichment is not ground truth.
- **Short-text false positives.** Generic descriptions (“prestations informatiques”) can match spuriously.
- **Structural right-censoring.** Contracts from 2021–2024 are largely censored; the positive `start_year` hazard partly reflects study truncation, not a genuine cohort trend.
- **Modest discrimination.** Concordance ≈ 0.65 — useful for prioritisation, not sufficient for high-stakes individual decisions.

6 Completed Work and Remaining Work

Completed. BOAMP acquisition and cleaning; buyer SIREN enrichment (2026-06-24); renewal-event reconstruction (705 events, 64.1%); survival modeling (KM, Cox, AFT) with sensitivity and robustness checks; operational 12/24-month risk indicators; rule-based technology taxonomy.

Remaining.

- **Manual validation of linked pairs.** The 150-row sample is ready; the review has not been executed. Until done, there are no independently verified precision/recall estimates for the linking step.
- **Trained NLP classifier.** The technology taxonomy is rule-based (CPV + keywords); the planned learned classifier was not implemented.
- **Change-point / trend detection (Phase 4).** Scoped but not started.

- **DECP semantic linking.** Applied to BOAMP only; DECP linking remains at the lexical prototype stage.
- **Live prediction scoring.** The current model evaluates contract age as of the study end date; a production run would re-evaluate each contract at a live prediction date.

7 Conclusion and Next Steps

The internship delivered a reproducible end-to-end pipeline: BOAMP notice acquisition → cleaning → buyer SIREN enrichment → renewal-event reconstruction → survival modeling → operational risk indicators. The main methodological result is lifting the renewal linking rate from 11.3% (lexical baseline) to 64.1% using Sentence-Transformer embeddings. Survival modeling shows renewal timing is driven by declared contract duration (median 48.2 months) rather than technology family, and the log-normal AFT model translates this into actionable 12/24-month renewal-risk rankings.

Immediate next steps (prioritised):

1. **Execute manual validation** of the 150-row sample to obtain independently verified precision/recall estimates for the linking step.
2. **Re-score contracts at a live prediction date** (current scores use end-of-study ages).
3. **Review medium/low SIREN matches** to extend buyer consolidation beyond the 660 HIGH-confidence rows already applied.
4. **Replace the rule-based taxonomy** with a trained text classifier.
5. **Implement change-point detection** (Phase 4).